

Chemical Equilibrium

The chemical equilibrium reactions Equation (D5.1) are modeled with chemical equilibrium constants (Bethke, 1996). The governing equations for the chemical equilibrium reactions are:

$$Q_j - K_{eq,j} = 0, \quad j = 1, \dots, R_{aq} \quad (D6.1)$$

with

$$Q_j = \prod_{i=1}^{n_{aq}} a_i^{v_{ij}} \quad (D6.2)$$

where $K_{eq,j}$ is the chemical equilibrium constant for the aqueous reaction j , a_i is the activity of component i , v_{ij} are the stoichiometry coefficients and Q_j is the activity product. Kharaka et al (1989), and Delany and Lundeen (1990), provide tables of values of $K_{eq,j}$ as a function of temperature for many aqueous reactions.

The activities a_i are related to the molality m_i (moles per kg of H₂O) as follows:

$$a_i = \gamma_i m_i, \quad i = 1, \dots, n_{aq} \quad (D6.3)$$

where γ_i is the activity coefficient. For ideal solution, $\gamma_i = 1$, and the activity is equal to the molality. However, for most cases, the solution is non-ideal and the preferred model for ionic activity coefficients is the B -dot model (Bethke, 1996):

$$\log \gamma_i = - \frac{A_\gamma z_i^2 \sqrt{I}}{1 + \dot{a}_i B_\gamma \sqrt{I}} + \dot{B} I \quad (D6.4)$$

where A_γ , B_γ and $\dot{B}$ are temperature dependent parameters, $\dot{a}_i$ is the ion size parameter. I is the ionic strength given by:

$$I = \frac{1}{2} \sum_{k=1}^{n_{aq}} m_k z_k^2 \quad (\text{D6.5})$$

with z_k being the charge of the k -th ion.

The activities of $\text{CO}_2(aq)$ and of H_2O can be assumed to be equal to 1 without introducing significant errors.